

Process Characterization Plan

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCP-006

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original, accurate and the associated data-integrity attributes; AMV, analytical method validation; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA,

critical quality attribute; Cpk, one-sided process capability index; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; PPQ, process performance qualification; PRESS, predicted residual error sum of squares; SDM, scale-down model; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TCID₅₀, 50 % tissue culture infectious dose; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Low-pH viral inactivation is the dedicated inactivation step for enveloped virus in the A-Mab drug substance process, and it is Step 6 of the train. It receives the Protein A eluate, which is already acidic, titrates it to the target pH, holds it under mixing for a defined time at a controlled temperature, and neutralises it for the cation exchange step that follows (Step 7). This plan defines the studies that will characterize the step, and it fixes the acceptance and decision criteria before any data are generated.

The plan covers the 4 process parameters of the low-pH hold and the 3 responses measured across them. Of those parameters, 3 enter a multivariate design and 1 is assessed one factor at a time, as assigned by the pre-characterization risk assessment (RA-001). All studies are executed on a qualified scale-down model (SDM) of the commercial hold vessel, and the clearance measurements use a separate small-scale virus spiking model. The work supports the technology transfer described in PTP-001, it forms part of the Stage 1 process design record expected under the lifecycle approach (U.S. Food and Drug Administration 2011), and the clearance element follows the viral safety framework of ICH Q5A(R2) (International Council for Harmonisation 2023a).

Three things sit outside this plan. Confirmation of the ranges at commercial scale belongs to Stage 2 and to the process performance qualification campaign (PPQ), not to characterization. The enveloped-virus clearance credited to anion exchange and to small-virus retentive filtration is characterized under PCP-008 and PCP-009, so only the contribution of this step is studied here. Host cell protein (HCP) precipitates during a low-pH hold and is removed by the depth filtration that follows the neutralisation, but the extent of that clearance is not predictable and no HCP claim will be made for this step (CMC Biotech Working Group 2009).

2 Objectives

The characterization has five objectives, and each one supports a decision the control strategy has to make.

- (i) Quantify the effect of inactivation pH, hold time and temperature on the inactivation of enveloped virus, measured as the XMuLV log reduction factor (LRF), over the ranges given in Section 6.1.
- (ii) Quantify the effect of the same three parameters on the two quality attributes the hold can change, which are aggregate content and acidic charge variants.
- (iii) Define the design space, which is the region inside the characterized space over which every response meets the criterion set for it in Section 7, and establish where the normal operating ranges sit inside it.
- (iv) Establish a proven acceptable range (PAR) for each multivariate parameter against each attribute the step governs, by the two analyses specified in Section 8. The acceptable range of A-Mab concentration follows from the univariate assessment
 - (1) against the same criteria.
- (v) Provide the evidence needed to classify every parameter under SOP-4001 and to state the contribution of the step to the control strategy.

Objectives (i) and (iii) carry the viral safety claim, which is the reason the step is characterized at all, and the remaining objectives bound the conditions under which that claim holds and turn it into something the control strategy can act on. Every objective is answered in PCR-006, which is the deliverable of this plan.

3 Related documents

This plan sits inside the A-Mab characterization package. Its parents, its paired report and the master report it rolls up into are given in Table 3.1.

Table 3.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-006	Process Characterization Report (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The controlled procedures and the validated analytical methods the studies depend on are given in Table 3.2. Operation of the step is governed by SOP-2009, qualification of the scale-down model by SOP-1001, and the design and analysis of the multivariate studies by SOP-1002.

Table 3.2: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step inactivates enveloped virus by acid denaturation. Exposure to low pH changes the conformation of the viral envelope glycoproteins irreversibly, so the particle can no longer fuse with a host cell membrane, and the loss of infectivity follows first-order kinetics in the duration of the exposure (International Council for Harmonisation 2023a). The rate constant rises as pH falls and as temperature rises, and the same two conditions act on the antibody, which is only marginally stable at low pH, so the step is a balance between the inactivation the process needs and the aggregation it has to avoid (CMC Biotech Working Group 2009).

Novacyte Biologics has operated a low-pH hold on three prior humanized IgG1 products (X-Mab, Y-Mab and Z-Mab), and the conditions of the hold have not changed materially across those products or through A-Mab clinical supply. The inactivation kinetics established on the prior products transfer directly to A-Mab, so the kinetic form of the step is settled platform knowledge before the first run is executed. Prior knowledge of that kind is what allows a characterization study to confirm and bound a known mechanism instead of establishing one.

Three expectations follow from the mechanism, and the study is designed to confirm or refute each of them. Inactivation should improve as pH falls and worsen as pH rises, so the high edge of the pH range is the direction of risk for viral safety. Aggregate content should rise with hold time and with temperature, because acid-induced partial unfolding is slow and thermally activated, which makes the long and warm corner of the design the direction of risk for product quality. Acidic charge variants should increase over the hold through deamidation of asparagine residues, and the increase is expected to be small against the width of the drug substance criterion.

The characterization ranges are wider than the ranges the commercial process will use, and each edge has a reason taken from the platform studies (CMC Biotech Working Group 2009). Inactivation pH is studied over 3.2–4 against a normal operating range (NOR) of 3.4–3.6. Earlier univariate work on the platform found that antibody precipitation can occur just below the low edge of that range, so it was not extended further. The high edge is the highest pH at which inactivation has been studied on the platform, and it lies well above the normal operating range, so that loss of inactivation at high pH is bounded inside the design. Hold time is studied over 60–180 min, which reaches well past the upper edge of its normal operating range. Its high edge is the longest hold the platform has carried without loss of acceptable product quality, and it leaves room for a hold extended by a downstream delay to fall inside the characterized space. Temperature is studied over 15–25 °C. The high edge sits above the normal

operating range to confirm that the antibody is stable there, since a warmer hold raises the rate of aggregation as well as the rate of inactivation. The low edge sits below the normal operating range to measure how much a colder hold slows the inactivation, which is the direction that carries the viral safety risk.

4.2 Quality attributes in scope

The step sets one critical quality attribute (CQA) and can change two others. They are given in Table 4.1 with the acceptance criterion and the criticality of each. Criticality follows the continuum approach of ICH Q9 (International Council for Harmonisation 2023b). It is assessed on two dimensions, which are the potential impact of the attribute on safety and efficacy and the uncertainty attached to that impact, and an attribute at the top of the scale (high or very high) is treated as critical (CMC Biotech Working Group 2009). The Tool #1 score in Table 4.1 is the product of the two dimensions, so it does not rank the three attributes in the order their criticality does. An attribute of very high impact whose behaviour is well established scores below one of moderate impact that is less well characterized.

Table 4.1: Quality attributes governed by the low-pH hold.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4

Enveloped-virus clearance is the attribute the step exists to deliver and it carries the highest criticality of the three. Its Tool #1 score is the middle one, because the consequence of a failure is severe but the mechanism is well understood and little uncertainty attaches to it. Aggregate content carries the highest score of the three, because its impact is the one assessed with the most uncertainty.

The acceptance criterion for enveloped-virus clearance is cumulative over the process, so no single step meets it alone. A-Mab claims enveloped-virus clearance at three steps, which are the low-pH hold, anion exchange (Step 8) and small-virus retentive filtration (Step 9), and the three mechanisms are independent of one another. The contribution required from this step follows from the cumulative requirement and from the clearance credited to the other two, and it is stated in Section 7.

Aggregate content is formed in the production bioreactor and reduced at cation exchange (PCP-003 and PCP-007). This step neither forms it in culture nor clears it, but it can add to it, and the study is designed to bound how much it adds across the characterized space. Acidic charge variants are of very low criticality and have a wide criterion, and they are measured here because deamidation is dependent on pH

and on time, and because this hold is the longest low-pH exposure in the purification train.

4.3 Risk-based prioritization of parameters

RA-001 assigned a study type to every parameter of this step before this plan was written. It scored each parameter on the severity of the attribute the parameter puts at risk, on how often a departure was expected to occur before the step was characterized, and on how likely such a departure was to be detected in process, and it combined the three into an initial risk priority number. The study type followed from whether the parameter was carried into the designed experiment for the step. Every parameter that acts on enveloped-virus clearance ranks at the top of that assessment, because the attribute is scored at the highest severity in the register and cannot be measured on a routine manufacturing batch.

Three parameters carried a multivariate assignment. Inactivation pH, hold time and temperature all enter the inactivation rate, they act on the product at the same time, and there is no prior basis for assuming that their effects stay additive across the full width of the characterization ranges. A design that varied them one at a time could neither detect that nor bound it.

A-Mab concentration carried a univariate assignment. RA-001 scored it against process performance rather than against a quality attribute of the step, which put it lowest of the four parameters, and it is not a factor of the designed experiment. Its normal operating range is the whole of its characterization range, because the concentration of the Protein A eluate is set upstream (PCP-005) and is not adjusted at this step.

Other operating variables of the step are fixed procedurally by the operation SOP (SOP-2009) and were not carried forward as characterization parameters. They include the identity and molarity of the titrant, the neutralisation target, the mixing configuration and the vessel material. Their control is verified in the batch record for each run, and a departure from any of them is handled as a deviation.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies are executed on a scale-down model of the commercial hold vessel, qualified under SOP-1001 before the first characterization run. A hold has no separation mechanism, so the model carries fewer scaling dependencies than a chromatography step, and the scale-independent variables (pH, temperature, hold time, protein concentration) can simply be operated at their commercial equivalents. The scale-dependent variables are vessel geometry, mixing time, titration rate and the ratio of surface area to volume, and they are set so that the titration and neutralisation profiles match the commercial records.

Qualification compares replicate runs at the set-point against the commercial-scale record for the same operation. The comparison covers the attributes of the load, the attributes of the pool at the end of the hold, and the time taken to reach the target pH on titration. The last of these matters more here than it would at a chromatography step, because the hold clock starts when the target pH is confirmed and any difference in titration time is a difference in exposure. The number of replicate runs, and the difference judged acceptable for each compared attribute, are fixed in the qualification protocol issued under SOP-1001 before the comparison is run. The acceptance criterion for qualification is given in Section 7.

Virus clearance is measured on a separate small-scale spiking model, which is what a clearance claim requires (International Council for Harmonisation 2023a). Live XMuLV is added to the load at a defined titre, and the spiked volume is held to a small fraction of the load volume so that the matrix of the material is not altered. Spiked material is handled in a laboratory that is physically separate from manufacturing and none of it returns to the process. The load for the spiking runs is Protein A eluate from a representative campaign, so that the properties that could change the outcome (buffer composition, protein concentration, impurity burden) are those the commercial step will see.

Centre-point replicates carry the pure-error term for the lack-of-fit tests described in Section 5.5. The screening design includes 3 centre points and the response-surface design includes 4, which give 2 and 3 degrees of freedom for pure error. Replication at the centre also gives the reproducibility of each response under nominally identical conditions, which PCR-006 reports as a coefficient of variation per response.

5.2 Operation

The step is operated as a batch hold in three phases. The Protein A eluate is titrated to the target pH with acid under mixing, and the pH is measured in the vessel after a defined mixing period. The batch is then held for the target time with the jacket controlling temperature, and the hold clock starts when the target pH has been confirmed. At the end of the hold the batch is neutralised to the load condition of the cation exchange step and filtered through a depth filter, which removes precipitated HCP and any product that came out of solution during the hold.

Set-points and normal operating ranges for the four parameters are given in Table 6.1. The buffers, the titrant and the neutralisation base are prepared and released to platform specifications under SOP-2009. They are identity and quality controlled, and they are not characterization parameters. The load is the Protein A eluate produced under PCP-005, and its attributes (concentration, aggregate content, HCP burden) are controlled at that step.

5.3 Analytical methods

Each response is measured by one validated method, and the three methods are given in Table 3.2. Enveloped-virus inactivation is measured as the XMuLV log reduction factor by infectivity titration (AMV-3017), read at the 50 % tissue culture infectious dose endpoint. The log reduction factor is the $\log_{10}$ of the total virus in the spiked load minus the $\log_{10}$ of the total virus recovered at the end of the hold. Both terms are total quantities, so the calculation accounts for the volume change on titration.

Aggregate content is measured by size exclusion chromatography (SEC-HPLC, AMV-3011) and reported as the percentage of high molecular weight species. Acidic charge variants are measured by imaged capillary isoelectric focusing (icIEF, AMV-3013) and reported as the percentage area of the acidic region. Both are platform methods already validated for A-Mab drug substance, so their performance is documented in the validation reports and is not reproduced in this plan.

Two properties of the infectivity assay bound what the study can conclude. Its intermediate precision is a standard deviation of 0.3 $\log_{10}$, which sets the smallest difference in log reduction factor the design can resolve and is the reason an effect is judged against the pure-error term of the design. Its limit of quantitation is 1.5 $\log_{10}$ TCID₅₀/mL, which together with the achievable spike titre sets the largest reduction the study can report. A condition that inactivates the spike below the limit of quantitation is reported as a reduction of at least the measurable value, and it is not extrapolated beyond it.

5.4 Sampling plan

Sampling is designed so that every run yields one infectivity result, one aggregate result and one charge-variant result, and so that the centre points also give the shape of the inactivation curve.

Each run is sampled at three points. The first sample is taken from the spiked load before titration and fixes the starting virus titre. The second is taken at the end of the hold, immediately before neutralisation, and carries the infectivity, aggregate and charge-variant results for that run. The third is taken after neutralisation and depth filtration and is retained as a reserve for investigation.

The centre-point runs are sampled during the hold as well, at 60 and 90 min. These samples confirm the shape of the inactivation curve without adding runs to the design, and they carry the evidence for the first-order assumption that any statement about hold time relies on. Product quality samples are not taken during the hold, because the aggregate response of the step is defined at the end of it.

Every sample is assayed inside the stability period established for its method. Sample identity, time of collection and storage condition are recorded in the batch record of the run, and a sample whose record is incomplete is not used in the analysis.

5.5 Statistical methods

All analysis follows the procedure for the design and analysis of designed experiments (SOP-1002). Each factor is coded so that the centre of its characterization range is zero and the two edges are -1 and $+1$, which puts the estimated effects on a common scale and makes them comparable across factors measured in different units.

The screening analysis fits all main effects and all two-factor interactions by ordinary least squares. An effect is reported as twice the coded coefficient, which is the change in the response between the two edges of the range. Significance is judged at $\alpha = 0.05$. The screening model has 7 terms in 11 runs, so it is close to saturated. It is used to identify which factors and which interactions are active, and it is not used as a predictive model.

The response-surface analysis fits the full quadratic model in the same factors, adding the pure quadratic terms so that curvature can be estimated. That model is the predictive model, and it is the basis of the design space declared in Section 7 and of the proven acceptable ranges derived in Section 8. Its adequacy is judged on the overall F test, on R^2 , on the adjusted R^2 , on the predicted R^2 computed from the predicted residual error sum of squares (PRESS), and on an analysis of variance that partitions the residual into lack of fit and pure error. With 18 runs and 10 terms the design leaves 8 residual degrees of freedom, of which 3 are pure error and 5 are lack of fit.

A factor with no significant effect on any response is not dropped from the record. It is reported as inactive over the range studied, it is classified on that basis, and it stays

in the knowledge space as evidence that the responses are robust to it. A null result of that kind is one of the more useful outcomes a characterization study can produce, because it removes a parameter from the list of things the control strategy has to hold tightly.

Commercial-scale capability is estimated by Monte-Carlo simulation of 2,000 batches, with every parameter drawn inside its normal operating range. Capability is reported as a one-sided Cpk against the governing limit, because each criterion for this step binds in one direction only. Run order within each design is randomised, and the randomisation is recorded before execution so that it cannot be adjusted afterwards.

6 Study design

6.1 Factors, ranges and study type

The design separates the three parameters that act on the inactivation rate from the one that does not. Table 6.1 gives the set-point, the characterization range, the normal operating range and the study type for each of the four parameters.

Table 6.1: Parameters, ranges and study type for the low-pH hold.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Inactivation pH	pH	3.5	3.2-4	3.4-3.6	multivariate
Hold time	min	90	60-180	60-120	multivariate
Temperature	°C	21	15-25	19-23	multivariate
A-Mab concentration	g/L	20	5-35	5-35	univariate

The characterization ranges are the edges of the knowledge space and the normal operating ranges are where the commercial process will run. Every characterization range covers the whole of its normal operating range and extends past it in the direction in which an excursion would carry risk, which is what makes these studies usable for assessing a deviation later in the life cycle. A-Mab concentration is the exception in the table. Its normal operating range is its whole characterization range, because the value is inherited from the Protein A eluate (PCP-005) and is not adjusted at this step.

The three multivariate factors are coded A to C, and that coding is used in every effect and coefficient table of PCR-006. The legend is given in Table 6.2.

Table 6.2: Factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

6.2 Screening design

The screening study is a two-level full factorial in the three multivariate factors, augmented with centre points. It has 8 factorial runs at the corners of the characterized space and 3 centre-point replicates, 11 runs in total. A full factorial estimates all three main effects and all three two-factor interactions without aliasing, which a fractional design in three factors could not do, and the run count is small enough that the fraction would save little.

The centre points serve two purposes. They estimate pure error, and they test for curvature, because a difference between the mean of the centre points and the mean of the factorial points is evidence that a first-order model is not sufficient. Curvature is expected at this step, since inactivation is exponential in time and pH acts on the rate constant. The response-surface study is therefore planned outright, and it is not made conditional on the outcome of the screening study.

The planned coded design matrix is given in Appendix A (Section 14).

6.3 Response-surface design

The response-surface study is a face-centred central composite design (CCD) in the same three factors. It has 8 factorial runs, 6 axial runs placed on the faces of the cube, and 4 centre-point replicates, 18 runs in total. Facing the axial points keeps every run inside the characterization ranges, which matters here because neither edge of the pH range can be moved outward, for the reasons given in Section 4. A design with axial points outside the cube would place runs below the precipitation edge and above the highest pH at which inactivation has been studied.

All three factors go forward from the screening design into the response surface. This is a design decision and not a result of the screening study, since the factor set is small enough that dropping one would save few runs, and each of the three is expected to be active on at least one response. If the screening study finds a factor inactive on every response, the response-surface design is executed as planned and the factor is reported as inactive on the evidence of both designs.

The planned coded design matrix is given in Appendix B (Section 15).

6.4 Univariate assessment

A-Mab concentration is assessed one factor at a time, at the low edge, the set-point and the high edge of 5–35 g/L, with pH, hold time and temperature held at their set-points. Each condition is spiked and assayed as a full run, so the univariate results are directly comparable with the centre points of the multivariate designs. Spiking every condition is deliberate. A higher antibody concentration has been reported to slow low-pH inactivation (CMC Biotech Working Group 2009), so the assessment has to carry the log reduction factor and not the product quality attributes alone. The

acceptable range of A-Mab concentration is established from this assessment, against the criteria in Table [7.1](#).

The reason for the univariate treatment is mechanistic, and it rests on two expectations the study will test. The rate of inactivation at low pH is set by the three multivariate factors (pH, temperature, duration of the exposure), and it is not expected to depend materially on the concentration of the antibody in which the virus is suspended. Aggregation is expected to depend on concentration, but through a mechanism that is not expected to interact with the hold conditions over the range studied. If either expectation is contradicted by the univariate results, the parameter is escalated to a multivariate study and this plan is amended before that study is executed.

The other operating variables named in Section [4.3](#) are not studied at all. They are held procedurally and their control is verified in the batch record of each run.

7 Acceptance and decision criteria

The study passes when three conditions hold together: the scale-down model is qualified, every response model is adequate, and the design space defined from those models contains the normal operating ranges of the parameters. Each of the three is defined below before execution, and none of them may be redefined after the data are seen.

Table 7.1 gives the criterion each response has to meet and the basis of each criterion.

Table 7.1: Acceptance criteria for the measured responses.

Re- sponse	Quality attribute	Accep- tance criterion	Unit	Basis
Aggre- gates (HMW, %)	Aggregates (HMW)	≤ 5	% HMW (SEC)	Drug-substance specification
XMuLV LRF (log ₁₀)	Viral clearance — XMuLV (enveloped)	≥ 4.93	log ₁₀ (this step)	Cumulative requirement less the clearance credited to the other steps
Acidic variants	Acidic charge variants (deamidation)	≤ 40	% (CEX/icIEF)	Drug-substance specification

Two of the three criteria are drug substance specifications and apply to the pool as it leaves the step. The third does not work that way. Enveloped-virus clearance is claimed cumulatively over three steps, so the criterion for this step is the cumulative requirement of 16.7 log₁₀ less the clearance credited to anion exchange and to small-virus retentive filtration, which leaves 4.93 log₁₀ for the low-pH hold. That figure is a floor and not a target. The step is expected to deliver more than the floor, and the margin above it is what allows the cumulative claim to survive a weak result at one of the three contributing steps.

The scale-down model is qualified when the difference between the two scales at every compared attribute falls inside the difference set for that attribute in the qualification protocol (Section 5.1). The criterion is an equivalence criterion and not a test for a significant difference, because a failure to find a difference over a small number of runs is not evidence that the two scales agree. A model that fails on any attribute is corrected and requalified, and characterization does not begin until it passes. The qualification report is issued under SOP-1001 and is referenced by PCR-006.

A response model is adequate when the overall F test is significant at $\alpha = 0.05$ and the lack-of-fit test against pure error is not significant at the same level. The predicted R^2 is reported alongside the adjusted R^2 , and a large gap between the two is treated as evidence of over-fitting even where the lack-of-fit test passes. No numerical threshold is set for R^2 in this protocol, because no such threshold is grounded in prior data for this step, and a value chosen for the occasion would carry no information. Where a model is adequate but the fitted surface is flat against the precision of the assay (Section 5.3), the parameter is reported as inactive over the range studied and not as a small effect.

The design space is declared as the set of conditions inside the characterized space at which every response in Table 7.1 is predicted to meet its criterion. It is declared from the response-surface models. The screening models identify which factors are active and they are not used to draw the boundary. The study passes on this criterion when the normal operating ranges of the three multivariate parameters lie inside the design space, and when the worst-case corners of the normal operating box lie inside it as well. For viral safety the worst case is the corner at the highest pH, the shortest hold and the lowest temperature. For product quality it is the corner at the lowest pH, the longest hold and the highest temperature. Both are evaluated explicitly and reported whether or not they pass.

Parameter classification follows SOP-4001 and the criticality continuum of ICH Q9 (International Council for Harmonisation 2023b). A parameter whose variation across its normal operating range moves a critical quality attribute measurably is a critical process parameter (CPP). It is classified as a well-controlled critical process parameter (WC-CPP) where the control system holds it comfortably inside the range that keeps the attribute within its criterion, and it is classified as a CPP without that qualification where the range is narrow relative to the control capability (CMC Biotech Working Group 2009). A parameter with no demonstrated effect on any attribute is a general process parameter (GPP). The classification is an output of the study and it is reported in PCR-006, so this plan states the decision rule and stops there.

Capability is reported for each attribute the step governs, as a one-sided Cpk from the simulation described in Section 5.5. No minimum Cpk is set as an acceptance criterion in this protocol. The capability figure is evidence about the margin the process holds at commercial scale, and it is weighed against the criticality of the attribute when the control strategy is defined in PCR-006 and consolidated in PCMR-001.

A failed criterion is not handled by widening the design space. If the design space does not contain a normal operating range, the range is narrowed or the set-point is moved, and the change is justified in PCR-006 against the same models. If a model is not adequate, the affected design is repeated once the cause is understood, and a deviation is raised and dispositioned before the repeat is executed. Any departure from this plan is recorded and assessed for its effect on the conclusions, whether or not it changed a result.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is a range of one parameter over which one quality attribute stays inside its acceptance criterion. The design space is the multivariate region described in Section 7. The two are different objects, and PCR-006 will report both, because a univariate range is what a batch record and an operator can work with, while the design space is what the filing claims. Both are drawn from the response-surface models, which carry the multivariate parameters only. The acceptable range of A-Mab concentration comes from the univariate assessment instead (1).

The acceptance basis is the same as in Table 7.1. Aggregate content and acidic charge variants are judged against the drug substance specification. The XMuLV LRF is judged against the step floor back-calculated from the cumulative requirement, and it is never judged against the cumulative figure itself, which no single step can meet.

Two analyses are run for each combination of multivariate parameter and governed attribute. The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range. That scan uses the fitted response-surface model on a grid of 201 points, and the proven acceptable range is the contiguous part of the scan, containing the operating point, over which the predicted response meets its criterion.

The second analysis propagates the normal operating ranges of the other parameters across a grid of 81 points. At each grid point the other parameters are drawn inside their normal operating ranges, the fitted model is evaluated, and model residual variation is added, for 2,000 draws per point. The proven acceptable range from this analysis is the part of the scan over which the 95 % predictive interval of the response stays inside the criterion. The whole interval has to clear the limit and not only the predicted mean, so this analysis is a robustness criterion, it is expected to give the narrower of the two ranges, and it is the one the control strategy uses.

Every proven acceptable range is reported as a two-panel figure, with the at-set-point analysis on the left and the propagated analysis on the right. The parameter is on the horizontal axis and the response on the vertical axis, the acceptance limit is drawn, the acceptable part of the parameter range is shaded green, and the set-point and the normal operating range are marked.

Where the two analyses agree across the whole characterization range, the entire studied range of that parameter is proven acceptable and is robust to co-variation of the other parameters inside their normal operating ranges. Where the propagated analysis is narrower, PCR-006 names the attribute that binds and states the margin remaining between the normal operating range and the edge of the proven acceptable range.

Where the whole characterization range is proven acceptable for every attribute, the binding constraint on the design space is the multivariate corner and not any univariate range, and PCR-006 states that explicitly instead of leaving the reader to infer it.

Both analyses are bounded by the characterization ranges and by the fitted models. Neither supports a claim about conditions outside the ranges studied. Neither replaces confirmation at commercial scale, which belongs to Stage 2.

9 Data management and integrity

Study records are created, reviewed and retained under the ALCOA+ principles, so that each record is attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available.

Raw data are captured in the validated laboratory systems that serve each method. The process conditions of each run are recorded in the electronic batch record of the scale-down system, and the pH, temperature and time traces are retained as the primary record of the hold. Infectivity results are transferred from the assay system into the study data set under second-person verification, because a titration endpoint is read rather than integrated.

The analysis data set is locked before any model is fitted and the lock is recorded. A change after the lock is made as a new version with the reason recorded, and the analysis is then repeated in full. Analysis scripts and their outputs are retained with the study record, so that every table in PCR-006 can be regenerated from the raw data by a reviewer who has only that record.

Access to the systems is role based. Audit trails are enabled and are reviewed as part of the study record review, and records are retained for the period the licence application requires.

10 Roles and responsibilities

The study is executed by Process Development and Manufacturing Science and Technology, with the clearance runs executed by the virology laboratory. Responsibilities are given in Table 10.1. Each function approves this plan before execution and reviews the report against it.

Table 10.1: Functions and their responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owns this plan, qualifies the scale-down model, executes the process runs and authors PCR-006
Virology laboratory	Executes the spiking runs, performs the infectivity assays and issues the clearance results
Analytical Development	Performs the size-variant and charge-variant testing and maintains the validated methods
Statistics	Reviews the designs before execution, performs the model fitting and reviews the reported inferences
Manufacturing Science	Confirms that the scale-down model and the studied ranges represent the receiving site
Quality Assurance	Approves this plan and the report, and dispositions any deviation from either

11 Deliverables and schedule

The deliverable of this plan is PCR-006, the process characterization report for the low-pH hold. The studies run in the four phases given in Table 11.1, in that order. No calendar dates are fixed here, because each phase begins only when the preceding one has been reviewed and released.

Table 11.1: Study phases and their outputs.

Phase	Activity	Output
1	Scale-down model qualification against the commercial-scale record	Qualification report under SOP-1001
2	Screening study, executed in randomised order	Screening data set and effect estimates
3	Response-surface study, executed in randomised order	Response-surface data set and fitted models
4	Univariate assessment, analysis, capability simulation and reporting	PCR-006, including the design space and the proven acceptable ranges

The report is issued into the characterization package and rolls up into PCMR-001. Where a phase produces a result that changes the design of a later phase, the change is documented before that phase begins.

12 Risks and assumptions

Four risks could invalidate part of the study, and each has a defined response. They are stated here so that the response is agreed before the risk materialises.

Titration at small scale reaches the target pH faster than at commercial scale, so a commercial batch spends longer at intermediate pH before its hold clock starts. The scale-down model would then understate the total acid exposure of the step, and with it the aggregate burden. The response is to match the titration profile during qualification, to record the time to target pH for every run, and to treat a run whose titration time falls outside the qualified profile as a deviation.

The virus stock adds its own matrix to the load, and that matrix can interfere with the infectivity assay or be cytotoxic at low dilution. The response is to hold the spike to a small fraction of the load volume, and to run cytotoxicity and interference controls with every assay, so that an apparent reduction caused by the assay is distinguishable from one caused by the hold.

The study needs Protein A eluate that represents commercial material. If the eluate available differs in concentration or in impurity burden, the aggregation responses will shift and the fitted surfaces will describe the wrong load. The response is to record the load attributes for every run and compare them against the release record of the eluate, and to invalidate the affected runs where the load is not representative.

Inactivation may be complete at most design points, in which case the log reduction factor is censored at the limit of quantitation and the models cannot estimate an effect from it. The response is to maximise the spike titre within the constraint on spike volume, and to treat a censored result as a bound rather than as a value in the analysis. The low-pH, long-hold corner of the design is the one most likely to be censored.

Three assumptions underlie the plan. The SDM represents the commercial hold once it is qualified under SOP-1001. The Protein A eluate used as load represents commercial material. The clearance credited to the other two enveloped-virus steps in the modular claim is achieved, and those two steps (PCP-008 and PCP-009) carry the evidence for it. If the third assumption fails, the floor for this step rises and the design space reported in PCR-006 is reassessed against the new floor.

13 Approvals

This plan takes effect when it has been approved by the functions listed in Table 1. Approval signifies that the design, the acceptance criteria and the planned analysis are agreed before execution starts. A change to any acceptance criterion after approval requires a revision of this plan and re-approval by the same functions.

14 Appendix A — Planned screening design matrix

The coded matrix below is the screening design as planned. Coded -1 and $+1$ are the low and the high edge of the characterization range of each factor, and 0 is the centre of that range. The response columns are absent because no data exist at the time of approval, and they are reported in PCR-006.

Table 14.1: Planned screening design matrix, coded.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	center	0	0	0
10	center	0	0	0
11	center	0	0	0

15 Appendix B — Planned response-surface design matrix

The coded matrix below is the response-surface design as planned, in the same coding as Table 14.1. The axial runs sit on the faces of the cube, so every planned run lies inside the characterization ranges given in Table 6.1.

Table 15.1: Planned response-surface design matrix, coded.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	axial	-1	0	0
10	axial	1	0	0
11	axial	0	-1	0
12	axial	0	1	0
13	axial	0	0	-1
14	axial	0	0	1
15	center	0	0	0
16	center	0	0	0
17	center	0	0	0
18	center	0	0	0

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